

Comparing Habitat Suitability Modeling Approaches for the Eastern Oyster in the Pensacola Bay System

M. Fazio

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1 Introduction

The eastern oyster (*Crassostrea virginica*) are viewed as a foundational species as they are directly responsible for creating and providing numerous ecosystem goods and services such as food and habitat production, shoreline stabilization and protection, water quality improvement, nutrient cycling, and carbon sequestration (Geselbracht et al. 2024; Grabowski et al. 2012; Barnes et al. 2007). Additionally, they provide numerous economic benefits such as supporting oyster fisheries, recreational harvesting, tourism, along with the associated employment that

goes along with those industries (Geselbracht et al. 2024; Peyre et al. 2021). In one study, Grabowski et al. (2012) estimated the average annual value to range from \$10,325 to \$99,421 per hectare (2011 Dollars).

In recent decades, oyster reefs have experienced substantial declines, with only an estimated 15% remaining worldwide, and 50 to 80% being lost within the Gulf of (America) alone (Pollack et al. 2012; Peyre et al. 2021). The decline is largely attributed to a combination of anthropogenic activities such as over harvesting, habitat alteration, sedimentation and water quality decline, disease and changing hydrology (Peyre et al. 2021; Geselbracht et al. 2024). In response to these sharp declines, many restoration efforts have been undertaken with varying strategies and success rates; recently, planning documents released by the Deepwater Horizon Natural resource Damage Assessment Trustees called for large-scale development of reef networks to increase resilience (Geselbracht et al. 2024).

1.1 Oyster decline in the Pensacola Bay System

On such system to experience oyster decline is the Pensacola Bay System (PBS). The PBS is a large estuarine system located in Escambia and Santa Rosa counties in northwest Florida. The system is comprised of multiple waterbody's including Pensacola Bay, Escambia Bay, Blackwater Bay, East Bay, and Santa Rosa Sound; and is influenced heavily by freshwater input from the Escambia, Blackwater and Yellow Rivers (Christensen et al. 1997). Historically, the PBS supported extensive oyster reefs that provided critical habitat for a variety of species and substantial economic value to the surrounding communities (Christensen et al. 1997). However, over the past several decades, the extent of oyster habitat within the PBS has declined significantly due to a combination of factors including over harvesting, habitat degradation, and water quality issues. With some estimates indicating a decline of 15,000ha to just 91.5 ha of oyster habitat between the years of 1883 and 2021 (Geselbracht et al. 2024). This decline has prompted efforts to restore oyster reefs within the PBS (Pensacola and Perdido Bays Estuary Program n.d.; The Nature Conservancy 2021), with multiple studies aiming to identify the most suitable areas within the estuary to focus restoration efforts (Geselbracht et al. 2024; Christensen et al. 1997; Pensacola and Perdido Bays Estuary Program n.d.).

With limited resources available for restoration, it is critical to identify areas where chances of oyster survival and long-term reef establishment are at their highest, and where poor site selection could ultimately lead to unsatisfactory results. As with any ecosystem, the environmental conditions within the PBS vary greatly in time and space, and such variations can have great impacts on oyster survival and establishment. Therefore, it is important to understand the environmental factors that drive oyster occurrence so that scientist and policy makers can make informed decisions regarding said restoration site selection (Pollack et al. 2012; Barnes et al. 2007; Geselbracht et al. 2024).

1.2 Environmental controls on Eastern oyster habitat

Oyster recruitment, survival, and reef establishment are influenced by complex interactions between numerous physical, chemical, and biological environmental factors. As such, determining the most influential factors is a multivariate problem that requires careful consideration of each factor and its interactions with others (Peyre et al. 2021; Pollack et al. 2012; Barnes et al. 2007; Christensen et al. 1997). To help narrow the focus, the most commonly cited environmental factors were examined in the literature and are summarized here.

Perhaps the most influential and widely reported environmental driver of oyster suitability is salinity. Representing an important environmental gradient within estuarine systems that influences physiology, recruitment, disease, predation, and survival. Other important characteristics often reported can be seen in **?@tbl-environmental-drivers** and include: dissolved oxygen, temperature, depth, turbidity, and bottom type (substrate) (Peyre et al. 2021; Pollack et al. 2012; Barnes et al. 2007; Geselbracht et al. 2024; Christensen et al. 1997).

Table 1: Common environmental drivers of Eastern Oyster habitat suitability.

Factor	Description	Sources
Salinity	Controls feeding, growth, reproduction, recruitment, disease, predation, and survival.	(Barnes et al. 2007; Birch et al. 2021; Christensen et al. 1997; Geselbracht et al. 2024; Peyre et al. 2021; Pollack et al. 2012)
Temperature	Regulates metabolism, filtration, growth, spawning, and larval development.	(Barnes et al. 2007; Christensen et al. 1997; Geselbracht et al. 2024; Peyre et al. 2021; Pollack et al. 2012)
Substrate type	Stable hard substrate supports larval attachment and reef persistence, whereas soft or rapidly accumulating sediments can bury or destabilize oysters.	(Barnes et al. 2007; Birch et al. 2021; Christensen et al. 1997; Geselbracht et al. 2024; Peyre et al. 2021)
Dissolved oxygen	Adequate oxygen supports respiration, feeding, and growth.	(Christensen et al. 1997; Geselbracht et al. 2024; Peyre et al. 2021; Pollack et al. 2012)
Hydrology / water movement	Water movement regulates salinity and transports larvae, oxygen, and food.	(Barnes et al. 2007; Geselbracht et al. 2024; Peyre et al. 2021)
Depth / bathymetry	Depth influences exposure, temperature, oxygen conditions, and restoration feasibility.	(Christensen et al. 1997; Geselbracht et al. 2024; Pollack et al. 2012)
Turbidity / sedimentation	High suspended-sediment loads can interfere with filtration and settlement. Sediment deposition can smother oyster reefs.	(Peyre et al. 2021; Pollack et al. 2012)
Food availability / chlorophyll a	Chlorophyll a is often used as a proxy for food availability.	(Geselbracht et al. 2024; Peyre et al. 2021)

It should be recognized that not all environmental factors are adequately spatially distributed or represented within the available datasets, and therefore careful consideration should be given when selecting the final dataset used for modeling habitat suitability (Peyre et al. 2021; Geselbracht et al. 2024; Christensen et al. 1997).

1.3 Species Distribution Modelling

Quantifying the relationships between the distribution of species and the species-environment interactions have a long history in ecological research, having been described as far back as the late 17th early 18th century in which researchers observed biological patterns in terms of their relationships with geographical and/or environmental gradients (Rushton et al. 2004; Elith and Leathwick 2009). Since then, the propagation of large-scale spatial datasets, the development and integration of GIS, increased access to statistical packages and computational power have allowed for much more development in modeling complex species-environment interactions (Johnson and Gillingham 2005). Out of such developments has arisen a technique known as Species Distribution Modeling (SDM), which can be loosely defined as numerical tools that combine observations of species occurrence or abundance with environmental estimates (Elith and Leathwick 2009). According to Elith and Leathwick (2009), SDMs can be largely classified within several families of models: expert-driven or heuristic, environmental-envelope, environmental-similarity, regression-based, Bayesian and hierarchical, machine-learning and data mining, and ensemble or multimodal. Additionally, the selection of an SDM approach largely depends on a multitude of factors such as: the study's objective, quality and quantity of available data, ecological characteristics of the species, spatial and temporal scale, whether input data includes absence data or presence-only data (Elith and Leathwick 2009). With so many choices of modeling technique and the driving factors behind that choice, deciding on one model can be quite difficult and the choice usually defaults to a standard technique depending on the subject matter.

Habitat Suitability Indices

One popular approach used in many studies involving the Eastern oyster throughout the gulf region is that of the expert-driven Habitat Suitability Index (Barnes et al. 2007; Pollack et al. 2012; Geselbracht et al. 2024; Christensen et al. 1997). HSIs are typically used in situations where quantitative data on species distributions is difficult to collect or non-existent, relying on expert-driven variable selection, meta-analysis, and classification methods that transform environmental characteristics into standardized estimates of habitat suitability (Barnes et al. 2007; Pollack et al. 2012; Geselbracht et al. 2024; Johnson and Gillingham 2005).

Within the PBS, HSIs have been used to develop GIS-based and estuary-specific HSMs that integrate biological, physical, chemical, and management-related variables in order to identify suitable restoration areas with the system. In the mid 1990's Christensen et al. (1997) developed a HSI for the eastern oyster in the PBS to demonstrate the model frameworks usefulness in illustrating species suitability in a system wide approach. Geselbracht et al.

(2024) then followed and further expanded on the approach incorporating management factors into the model, adding a layer of regulatory influence to the model output versus the purely environmental analysis of the prior. Additionally, the Pensacola and Perdido Bay Estuary Program recently built on-top of Geselbracht et al. (2024) by incorporating some of their own data and refining some of the weighting (Pensacola and Perdido Bays Estuary Program n.d.).

These previous studies alone highlight the strengths and weakness of the HSI approaches. Some of the strengths include interoperability, adaptability, and the integration of ecological knowledge, but the results entirely rely on expert decisions regarding variable selection, suitability functions, weighting and scoring, and how to aggregate the results. All of these present opportunity for differing outputs even if the same inputs were used by different groups (Geselbracht et al. 2024; Johnson and Gillingham 2005).

Maximum Entropy Modelling (MaxEnt)

MaxEnt is a presence-background SDM framework based on general-purpose machine learning methods that are useful for making predictions using incomplete information by relating environmental conditions at known occurrence locations to the environmental conditions throughout the study area (Phillips et al. 2006; Elith et al. 2011). Some of the main benefits of MaxEnt is that it uses presence-only data, reducing the need for the often un-available absence data, it has the ability to model complex interactions between environmental conditions, and it has demonstrated high predictive performance while limiting over fitting through regularization (Phillips et al. 2009; Merow et al. 2013; Elith et al. 2011). An added benefit to this particular study is the fact that MaxEnt is tailored towards presence-only species occurrence data, making it highly suitable for the data set used here in which the FWC-FWRI (2026) datasets explicitly collected occurrence data without including absence locations.

However, the use of such algorithms should not be made blindly as many issues can arise if careful precautions are not taken. Due to the nature of ecological many biases in the data samples can arise such as spatial and temporal autocorrelation, group dependence structures, hierarchical structures and multicollinearity leading to overfitting model parameters and over-optimistic prediction confidence (**Dormann2013?**; Kass et al. 2021; Roberts et al. 2017). Thankfully, significant work has been put into this through cross-validation techniques and the development of model evaluation software packages such as ENMeval 2.0 implemented through the R statistical language (Kass et al. 2021; Roberts et al. 2017)

TODO: THINK OF BETTER SECTION NAME *Hypothesis*

To demonstrate how a MaxEnt SDM can be used as a complimentary tool to help researchers and practitioners leading ecological restoration activities, this study aims to develop an empirically fitted MaxEnt habitat-suitability model using observed oyster occurrence data and known environmental conditions. The primary objective will be to develop a spatially explicit SDM representing relative Eastern oyster habitat suitability throughout the PBS. Next, the resulting suitability surface will be evaluated against the existing Pensacola Bay HSI developed by Geselbracht et al. (2024) and iterated upon by Pensacola and Perdido Bays Estuary Program (n.d.). To achieve this, environmental predictors that are commonly associated with oyster

habitat suitability will be identified and obtained. The predictors will be used to fit the MaxEnt model and model performance will be evaluated using spatially separated cross-validation. Finally, we will compare the predicted surface to the existing HSI to determine areas of agreement and disagreement, and we will use these results to determine whether the MaxEnt framework provides an empirical alternative or complement to HSI-based restoration planning. It is hypothesized that the MaxEnt model will successfully identify patterns of habitat suitability for the Eastern oyster within the PBS. Overall agreement between the HSI and MaxEnt is expected to be high. However, MaxEnt is expected to produce a smoother, more continuous suitability surface because of the empirically fitted environmental relationships as opposed to predefined, lower-resolution HSI suitability envelopes. As a result, the MaxEnt model could prove beneficial to help narrow down project locations to help reinforce expert-driven model outputs.

2 Methods

2.1 Data Processing and Analysis

Software and Packages

All data processing, analysis, and modeling will be conducted using free open source software and packages to maximize accessibility and reproducibility. The majority of data manipulation and processing will be performed in the R programming language using the tidyverse suite of packages (Wickham et al. 2019; R Core Team 2024). Any data operations that need to be “spatially aware” will be performed using the Simple Features package (Pebesma 2018).

Analysis will be scripted and ran in Positron, an open-source integrated development environment developed by Posit that supports R, Python, scripts, notebooks, integrated consoles, along with interactive plots and data viewers.

Study Area

The study area for this project is the Pensacola Bay System which includes portions of Pensacola Bay, Escambia Bay, East Bay and Blackwater Bay. Boundaries of this study area were achieved by accessing Florida Department of Protections (FDEP) Waterbody IDs feature layer hosted on ArcGIS Online (FDEP 2005). ArcGIS allows direct access to hosted feature services via several formats. To avoid any potential compatibility issues, the GeoJSON format was chosen as this format is non-proprietary. Once retrieved, the dataset was filtered to include only relevant WBIDs (WBID = 548) and then combined to create a singular study area polygon. Creating the study area polygon is an important first step as it will be needed to extract relevant environmental data from much larger datasets in the coming data gathering.

Water Quality Portal Data

Water quality data is widely available for the entire United States via the EPA’s Water Quality Portal (WQP) which allows for a wide range of data querying for any data as it relates to Water Quality Monitoring programs (USGS (2026)). For this analysis, site data for Escambia and Santa Rosa County was queried using the WQP API then the study area polygon was used to extract only the relevant sites:

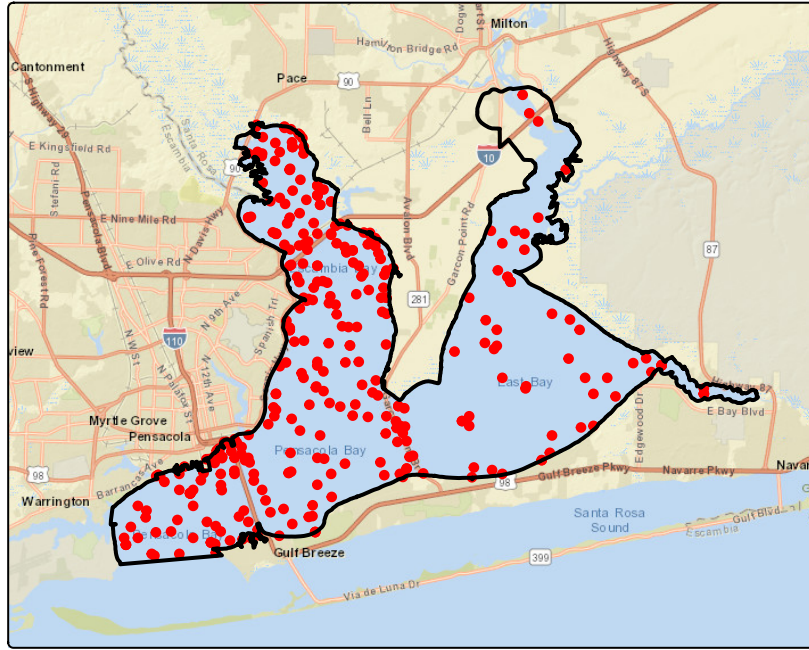


Figure 1: Pensacola Bay System with monitoring sites

Water Quality Data

Water quality data was selected based off of availability and relevance within literature. Each parameter was filtered to include just those measurements

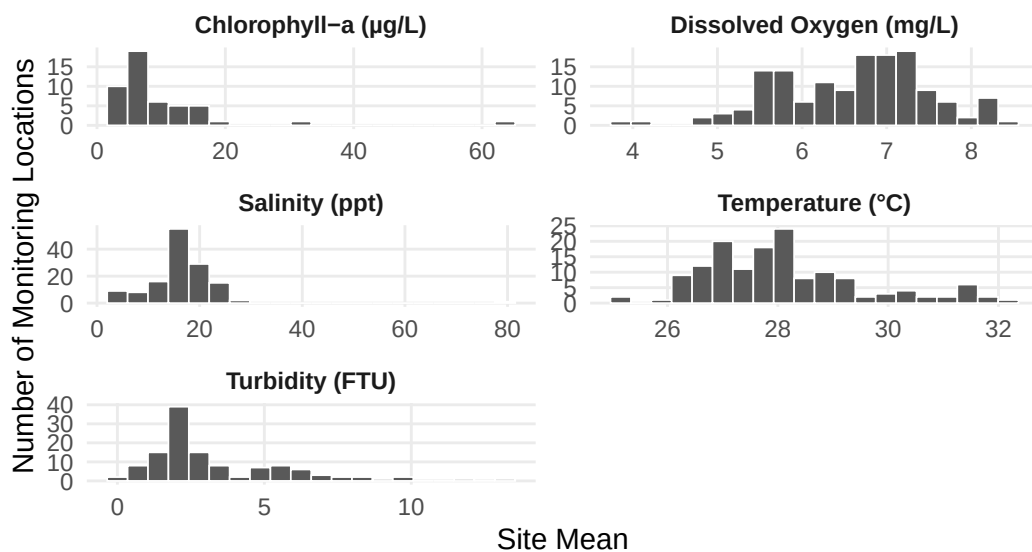
Table 2: Preliminary Environmental Variables Considered for the Eastern Oyster SDM

Variable	Obtained? (Y/N)	Data Type	Total Records	Data Source
Salinity	Y	Points	6994	National Water Quality Monitoring Council (2026)
Water temperature	Y	Points	7161	USGS (2026)
Substrate / bottom type	Y	Raster	NA	(ppbep?)

Variable	Obtained? (Y/N)	Data Type	Total Records	Data Source
Water depth / bathymetry	N	Raster	National Centers for Environmental Information (2018)	NA
Dissolved oxygen	Y	Points	6261	USGS (2026)
Chlorophyll-a	Y	Points	1847	USGS (2026)
Oyster bed locations	NA	NA	FWC-FWRI (2026)	NA

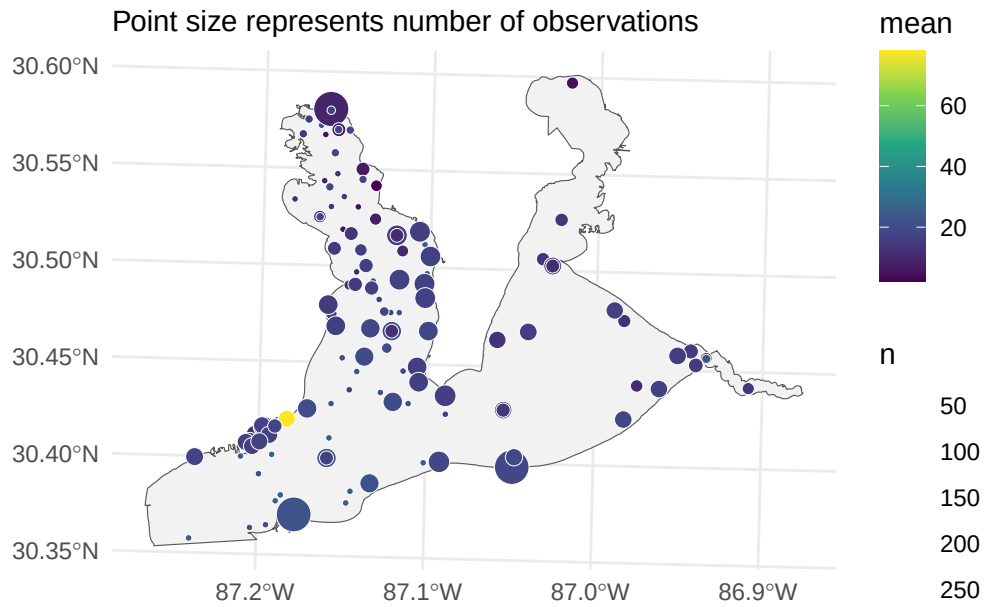
Distribution of Warm-Season Water Quality Conditions

Site-level means for monitoring locations in the Pensacola Bay System



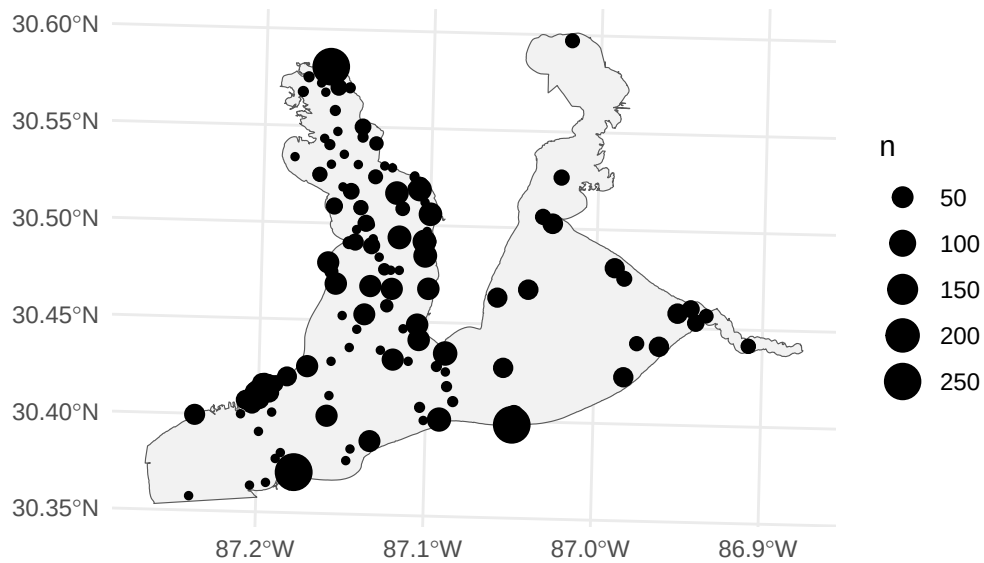
Warm-Season Salinity Monitoring Locations

Point size represents number of observations



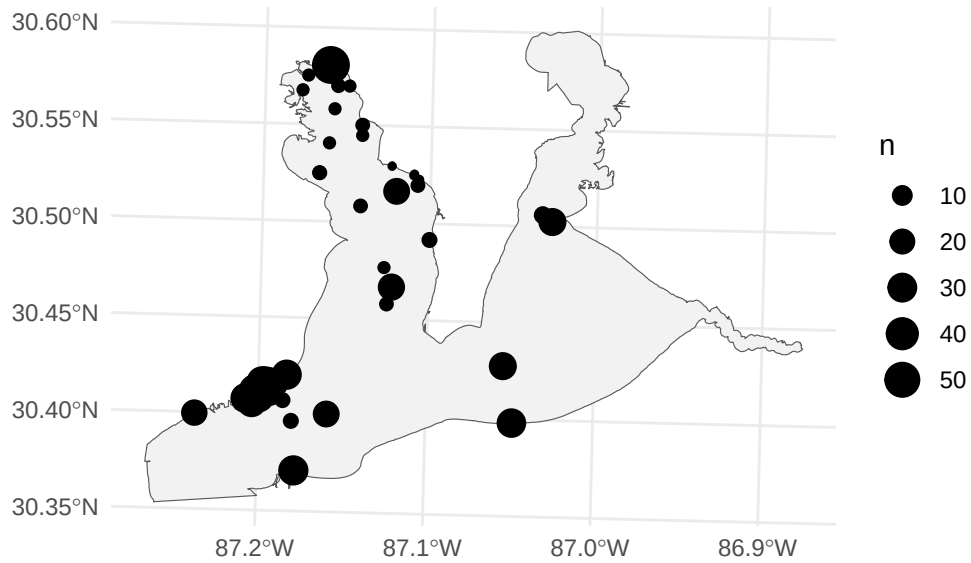
Warm-Season Temperature Monitoring Locations

Point size represents number of observations



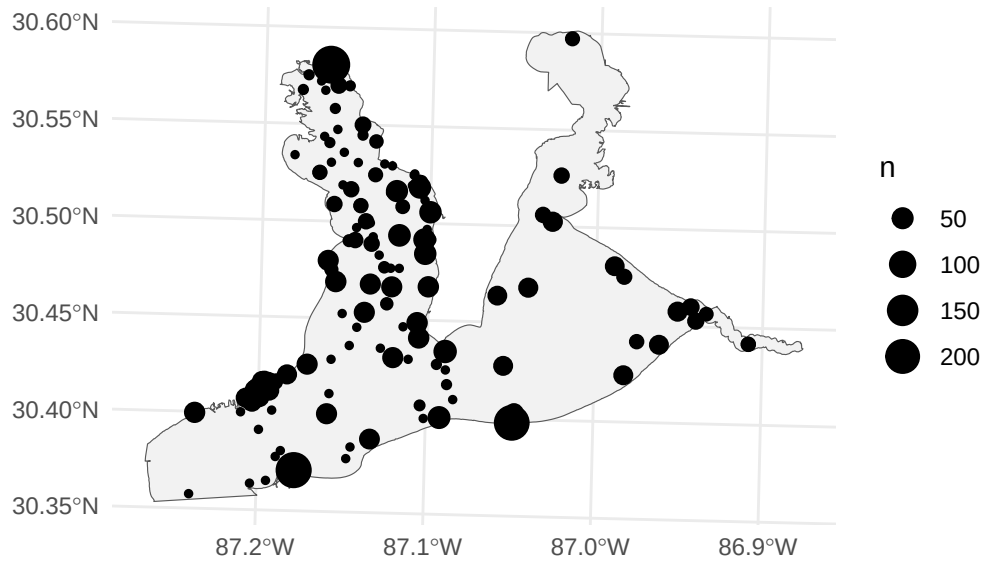
Warm-Season Chlorophyll-a Monitoring Locations

Point size represents number of observations



Warm-Season Dissolved Oxygen Monitoring Locations

Point size represents number of observations



Warm-Season Turbidity Monitoring Locations

Point size represents number of observations

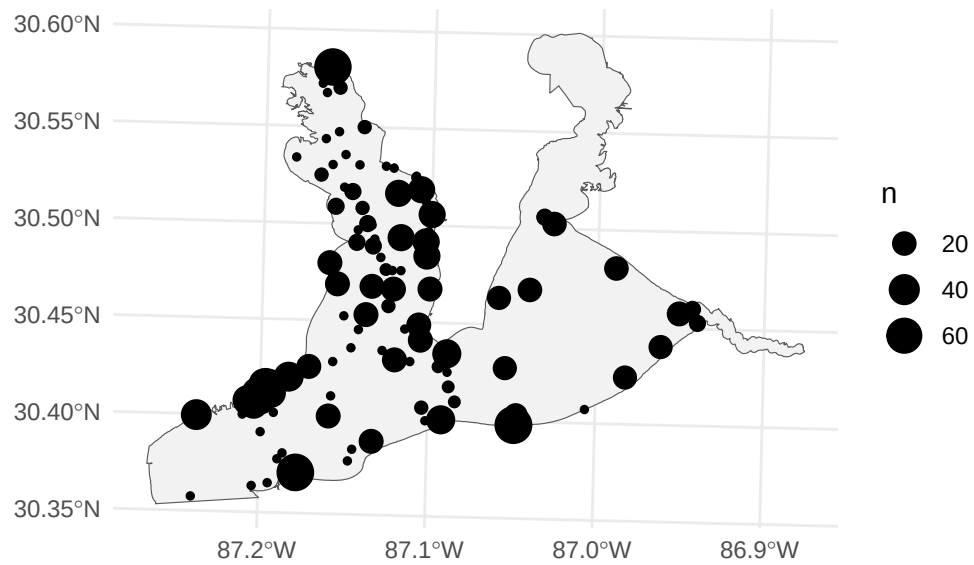


Table 3: Summary Statistics for Warm-Season Water Quality Data

Parameter	Sta- tions	Observa- tions	Mean	SD	Min	Q1	Me- dian	Q3	Max	First Sample	Last Sample
Chlorophyll-a ($\mu\text{g/L}$)	48	754	9.74	9.70	3.39	5.14	6.21	11.79	63.80	2005-05-03	2024-10-17
Dissolved Oxygen (mg/L)	145	3072	6.59	0.87	3.90	5.82	6.75	7.21	8.48	2005-05-03	2024-10-17
Salinity (ppt)	135	3401	16.55	7.35	2.31	14.02	16.54	18.91	77.85	2005-05-03	2024-10-17
Temperature ($^{\circ}\text{C}$)	145	3442	28.11	1.43	25.01	27.05	27.92	28.74	32.04	2005-05-03	2024-10-17
Turbidity (FTU)	122	1563	3.33	2.41	0.03	1.85	2.35	4.65	13.22	2005-05-10	2024-10-17

Spatial interpolation

Points are interpolated into prediction rasters using inverse path distance weighting (IPDW) (package `idpw`)....

3 Results

In progress...

4 Discussion

In progress...

5 Conclusion

In progress...

6 References

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